CHANDU PRASAD BHAIRAPU



AI Engineer • LLMs, Agents & AI Automation

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ABOUT ME

AI engineer with an MSc in Artificial Intelligence and 3 years of professional experience. I build LLM solutions end to end in Python: agents with LangGraph and MCP, the REST APIs and SQL databases around them, and the deployment with Docker on AWS and Azure, including GitLab CI/CD. I test what they deliver against ground truth and improve them on quality, cost and latency.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Built multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) and tested it systematically against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Sole developer owning an LLM-based knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream instead of an LLM approach, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed a Model Context Protocol (MCP) query service that makes the knowledge graph available to AI agents and Claude Code with millisecond latency.
- Developed Expert Interface, an internal web tool (React frontend, Python microservices backend) on Azure that lets domain experts review and approve knowledge-graph changes themselves.
- **Tech stack: Python, SQL (Postgres), LLM APIs, LiteLLM, LangGraph, LangChain, MCP, Claude Code, RAG, GraphRAG, FastAPI, React, Neo4j, MongoDB, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed an agentic AI application with LangGraph that routes requests across vector database queries, general inference, and summarization, and deployed it on AWS.
- Built the full stack: REST API backend with Python and FastAPI and a React frontend; API testing with Postman.
- Containerized the stack with Docker and deployed it to AWS using CDK; version-controlled in GitLab with a CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, REST APIs, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data.
- Built a Streamlit dashboard for chemists to simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Pandas, Scikit-Learn, XGBoost, TensorFlow/Keras, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy on unseen data.

PROJECTS

Project: Shop Support Agent | Python | Claude API | MCP | Shopify API | BigQuery

- AI agent that drafts replies to customer service tickets (order status, returns) from live order data in the Shopify Admin API, exposed to the agent through an MCP server, with a human approval step before anything is sent.
- Scheduled Python job loads orders and tickets into a BigQuery data warehouse and builds a weekly support report; a labelled ticket set tests reply quality, cost per ticket, and latency in a CI/CD pipeline, packaged with Docker for Google Cloud Run.

TECHNICAL SKILLSET

- **Programming & APIs:** Python, SQL, REST APIs (FastAPI), Git, Postman, React (beginner), Streamlit
- **LLMs & AI Tooling:** LLM APIs, LiteLLM, AI agents (LangGraph, LangChain), MCP, prompt engineering, Claude Code (custom skills), RAG, GraphRAG, AI output evaluation (RAGAS, PytrecEval)
- **Cloud & Deployment:** AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3), Azure, Docker, Kubernetes, GitLab CI/CD
- **Databases:** SQL (Postgres), NoSQL (MongoDB, DynamoDB), Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis
- **Ways of working:** Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - The Complete Neural Networks Boot camp

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)